

SKILL-7

INPUT



```
GNU nano 2.9.3 process_monitor.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Process Synchronization and Child Process Monitoring\n");
    printf("-----\n\n");

    printf("PATH Variable:\n");
    printf("%s\n", getenv("PATH"));

    printf("Enter command to search: ");
    scanf("%99s", command);

    printf("\nSearching for command: %s\n", command);

    char path[100];
    sprintf(path, sizeof(path), "/usr/bin/%s", command);

    if (access(path, X_OK) == 0)
    {
        printf("Executable found: %s\n\n", path);
    }
    else
    {
        printf("Executable not found.\n");
        return 1;
    }

    printf("Creating child process...\n");

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Executing %s command...\n", command);

        execl(path, command, (char *)NULL);

        perror("execl failed");
        exit(1);
    }
    else
    {
        printf("Parent Process\n");
        printf("Child PID: %d\n", pid);
        printf("Waiting for child using waitpid()...\n");

        int status;
        waitpid(pid, &status, 0);

        printf("Child Process\n");
        printf("Child PID: %d\n", pid);

        if (WIFEXITED(status))
        {
            printf("Child exited normally with status %d\n",
                WEXITSTATUS(status));
        }

        printf("Child terminated normally.\n");
    }

    return 0;
}
```

Read 78 lines

Help	Write Out	Where Is	Cut	Execute	Location	Undo
Exit	Read File	Replace	Paste	Justify	Go To Line	Redo

OUTPUT

```
administrator@VBCO-LAP-29 x + v - □ ×
administrator@VBCO-LAP-29:~$ mkdir skill-7
administrator@VBCO-LAP-29:~$ cd skill-7
administrator@VBCO-LAP-29:~/skill-7$ pwd
/home/administrator/skill-7
administrator@VBCO-LAP-29:~/skill-7$ nano process_monitor.c
administrator@VBCO-LAP-29:~/skill-7$ ls
process_monitor.c
administrator@VBCO-LAP-29:~/skill-7$ gcc -Wall -Wextra process_monitor.c -o process_monitor
administrator@VBCO-LAP-29:~/skill-7$ ls -l
total 24
-rwxr-xr-x 1 administrator administrator 16472 Sep 27 16:14 process_monitor
-rw-r--r-- 1 administrator administrator 1651 Sep 27 16:13 process_monitor.c
administrator@VBCO-LAP-29:~/skill-7$ ./process_monitor
Process Synchronization and Child Process Monitoring
=====

PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/
l/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files
/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/s
ystem32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0/
/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Pr
ogramData/chocolatey/bin:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Python312/Scripts/
:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Python312:/mnt/c/Users/Administrator/AppD
ata/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/Administrator/AppData/Local/Programs/Pytho
n/Python314:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/Adminis
trator/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Administrator/AppData/Local/GitHubDesktop/bi
n:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Users/Administrator/AppData/Local/Programs/Microsoft V
S Code/bin:/mnt/c/Users/Administrator/AppData/Roaming/npm:/snap/bin

Enter command to search: ls

Searching for command: ls
Executable found: /usr/bin/ls

Creating child process...
Parent Process
Child PID: 860
Waiting for child using waitpid()...
Child Process
Child PID: 860
Executing ls command...
process_monitor process_monitor.c
Child Process
Child PID: 860
Child exited normally with status 0
Child terminated normally.
```

administrator@VBCO-LAP-29

Child Process

Child PID: 860

Child exited normally with status 0

Child terminated normally.

administrator@VBCO-LAP-29:~/skill-7\$ ls -l

total 24

-rwxr-xr-x 1 administrator administrator 16472 Sep 27 16:14 process_monitor

-rw-r--r-- 1 administrator administrator 1651 Sep 27 16:13 process_monitor.c

administrator@VBCO-LAP-29:~/skill-7\$ gcc -Wall -Wextra process_monitor.c -o process_monitor

administrator@VBCO-LAP-29:~/skill-7\$./process_monitor

Process Synchronization and Child Process Monitoring

=====

PATH Variable:

/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Python312/Scripts:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Python312:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Python314:/mnt/c/Users/Administrator/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/Administrator/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Administrator/AppData/Local/GitHubDesktop/bin:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Users/Administrator/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Administrator/AppData/Roaming/npm:/snap/bin

Enter command to search: ls

Searching for command: ls

Executable found: /usr/bin/ls

Creating child process...

Parent Process

Child PID: 900

Waiting for child using waitpid()...

Child Process

Child PID: 900

Executing ls command...

process_monitor process_monitor.c

Child Process

Child PID: 900

Child exited normally with status 0

Child terminated normally.

administrator@VBCO-LAP-29:~/skill-7\$ ls -l

total 24

-rwxr-xr-x 1 administrator administrator 16472 Sep 27 16:16 process_monitor

-rw-r--r-- 1 administrator administrator 1651 Sep 27 16:13 process_monitor.c

administrator@VBCO-LAP-29:~/skill-7\$ nano process_monitor.c

administrator@VBCO-LAP-29:~/skill-7\$